

1. Administrative & Study Identification

Full Study Title: Study of Belzutifan (MK-6482) Plus Fulvestrant for ER+/HER2- Metastatic Breast Cancer (MK-6482-029/LITESPARK-029) **Short Title/Acronym:** LITESPARK-029 **Protocol Number:** MK-6482-029 **NCT Number:** NCT06428396 **Posting ID:** 183624459737160007 **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Belzutifan (MK-6482) and Fulvestrant **Phase of Development:** PHASE2 (Phase ID: PHASE2) **Trial Status:** RECRUITING (Status Name: RECRUITING | Status Description: Currently recruiting participants) **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy (measured by Progression-Free Survival [PFS]) of belzutifan plus fulvestrant compared to everolimus plus endocrine therapy (ET) in adults with estrogen receptor-positive, human epidermal growth factor receptor 2-negative (ER+/HER2-) unresectable metastatic breast cancer. **Secondary Objectives:** To evaluate the PFS rate at 6 and 12 months, Overall Survival (OS), Objective Response Rate (ORR), Clinical Benefit Rate (CBR) for ≥ 24 weeks, and overall safety and tolerability of the combination regimens.

2.2 Background & Rationale Endocrine-based therapy (ET), with or without cyclin-dependent kinase 4/6 inhibitors (CDK4/6i), is the standard of care for ER+/HER2- breast cancer. After disease progression, subsequent therapy options provide limited PFS gains due to resistance mechanisms. The transcription factor hypoxia-inducible factor 2 α (HIF-2 α) regulates key components of angiogenesis and subsequent metastasis. Belzutifan, an HIF-2 α inhibitor, has shown preclinical promise in suppressing tumor growth when combined with fulvestrant, providing a strong rationale to investigate this combination in the advanced/metastatic setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Everolimus + investigator's choice of fulvestrant or exemestane) **Blinding:** Open-label **Randomization:** Randomized (Parallel Assignment) **Phase Criteria:** Trial must be in PHASE2 (Criteria Type: PHASE_REQUIREMENT)

3.2 Planned Enrollment & Duration Targeted Enrollment: 120 evaluable participants **Number of Sites:** Multicenter (Global)
Estimated Study Start: 27-Nov-2024 **Estimated Primary Completion:** 05-May-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a diagnosis of ER+/HER2- invasive breast carcinoma that is either locally advanced disease not amenable to resection or metastatic disease not treatable with curative intent. 2. Documented radiographic confirmation of disease progression during or after the last administered endocrine therapy (ET) in the noncurative setting (e.g., progression on ≥ 12 months of ET + CDK4/6i therapy or received ≥ 2 lines of ET including CDK4/6i discontinued due to intolerance). 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 assessed within 7 days of randomization. 4. Provides additional tissue from the same sample used to determine ER and HER2 status locally. 5. Participants with hepatitis B surface antigen (HBsAg) positive are eligible if treated with antiviral therapy for ≥ 4 weeks and have undetectable viral load.

4.2 Exclusion Criteria 1. Breast cancer amenable to treatment with curative intent. 2. Inability to receive any of the endocrine therapies (fulvestrant or exemestane), inability to swallow oral medications, or uncontrolled nausea/vomiting/malabsorption. 3. Clinically significant cardiac disease (e.g., unstable angina, acute myocardial infarction within 6 months, or NYHA Class III/IV heart failure). 4. Known germline BRCA mutation and prior treatment with PARP inhibitors. Prior fulvestrant in any setting or prior cytotoxic chemotherapy/PARP inhibitor in the metastatic setting. 5. Symptomatic visceral spread at risk of rapidly evolving into life-threatening complications, active bleeding diathesis, or active pneumonitis/ILD. 6. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Experimental Arm:** Belzutifan 120 mg orally once daily (QD) + Fulvestrant 500 mg via intramuscular (IM) injection on Days 1 and 15 of Cycle 1, and Day 1 of each 28-day cycle thereafter. * **Active Comparator Arm:** Everolimus 10 mg orally QD + investigator's choice of either Fulvestrant 500 mg IM (schedule as above) OR Exemestane 25 mg orally QD. * **Comparator Drug Profile (Everolimus):** Drug Name: everolimus; Drug Preferred Name: everolimus; Trade Names: Zortress, Afinitor; ATC Code: L01EG02; EMA URL: https://www.ema.europa.eu/en/documents/product-information/afinitor-epar-product-information_en.pdf; Source

Level: 5. * **Duration of Treatment:** Administered in 28-day cycles until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, antiemetics, and hormone replacement therapy for adequately treated endocrine-related adverse events. **Prohibited:** Prior chemotherapy, PARP inhibitors in the noncurative setting, or strong inhibitors/inducers of CYP3A4.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Core Biopsy, Baseline Imaging
Baseline	Day 0	Randomization, First Dose Administration
Interim	Q8W to Q12W	Tumor Imaging (Q8W through week 56; Q12W thereafter), Safety Labs, Efficacy Assessment
End of Study	Upon PD / Toxicity	Treatment Discontinuation, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization to the first documented disease progression per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) based on Blinded Independent Central Review (BICR) or death due to any cause, whichever occurs first.

7.2 Secondary & Exploratory Endpoints * Progression-free Survival (PFS) at 6 months. * Progression-free Survival (PFS) at 12 months. * Overall Survival (OS). * Objective Response Rate (ORR) and Clinical Benefit Rate (CBR) per RECIST 1.1 by BICR.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical assessments, vital signs, ECG, and laboratory tests throughout the treatment cycles. **Serious Adverse Events (SAEs):** Standard reporting timelines (within 24 hours of investigator awareness) to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Safety data will be periodically evaluated by an internal/external monitoring committee to ensure subject safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy evaluations. Safety Set for safety endpoints. **Hypothesis Testing:** There is no formal hypothesis testing in this study. **Sample Size Justification:** The study targets 120 evaluable participants to estimate the PFS rate via the Kaplan-Meier approach with an event-driven cut-off date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits to ensure protocol adherence, data integrity, and patient safety. **Audit Trail:** Electronic Case Report Form (eCRF) capture with audit trails, data clarification forms, and blinded independent reviews for imaging endpoints.

11. Study Identifiers & Governance

Primary ID: MK-6482-029 **Registry Number:** NCT06428396 **Posting ID:** 183624459737160007 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** LITESPARK-029 **Official Regulatory Title:** A Phase 2, Randomized, Active-controlled, Open-label, Multicenter Study of Belzutifan Plus Fulvestrant in Participants With Estrogen Receptor Positive, HER2 Negative Unresectable Locally Advanced or Metastatic Breast Cancer After Progression on Previous Endocrine Therapy (LITESPARK-029)

12. Clinical Framework (Oncology Specific)

Disease Areas / Condition Name: Metastatic Breast Cancer **Preferred Name:** Tubular breast cancer metastatic **Preferred UMLS Name:** Estrogen Receptor-Positive HER2-Negative Metastatic Breast Cancer **Semantic Type:** Neoplastic Process **Diagnosis Definition:** Metastatic breast cancer, also known as stage IV breast cancer, is a condition where breast cancer cells spread from the original tumor in the breast to distant parts of the body, such as the bones, lungs, liver, or brain. Despite spreading to new locations, the cancer cells found in these organs are still breast cancer cells, and not cancer of the organ they have colonized. **Terminology / Ontology Codes:** * **MeSH Code:** D016629 * **SNOMED ID:** 254837009 * **SNOMED Hierarchy:** 424163001|Secondary malignant neoplastic disease (disorder)|Secondary malignant neoplastic disease (disorder) **Medical Methodology:** Targeted Therapy + Endocrine Therapy **Optimization Target:** HIF-2 α pathway inhibition combined with estrogen receptor degradation

13. Study Procedures & Methodology

Management Pathway: Standard oncological RECIST 1.1 assessment pathway
Education Protocol (HCP): Protocol specific training regarding safety monitoring and imaging acquisition
Education Protocol (Patient): Informed consent process, oral medication adherence training, and AE reporting guidelines
Quality Audit Mechanism: Blinded Independent Central Review (BICR) for primary imaging endpoints

14. Observation & Follow-Up Schedule

Total Duration: Event-driven progression monitoring
Onsite Visit Cadence: Days 1 and 15 of Cycle 1, and Day 1 of each subsequent 28-day cycle
Remote Monitoring Frequency: Periodic OS/survival follow-up post-progression
Checklist Review Interval: Tumor imaging every 8 weeks (Q8W) through week 56, and every 12 weeks (Q12W) thereafter

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				